of ions, hormones, lipids and assists in immune function. Both **Prothrombin** and **Fibrinogen are** involved in blood clotting. **Organic constituents** include urea, amino acids, glucose, fats and vitamins and the **inorganic constituents** include chlorides, carbonates and phosphates of potassium, sodium, calcium and magnesium.

The composition of plasma is not always constant. Immediately after a meal, the blood in the hepatic portal vein has a very high concentration of glucose as it is transporting glucose from the intestine to the liver where it is stored. The concentration of the glucose in the blood gradually falls after sometime as most of the glucose is absorbed. If too much of protein is consumed, the body cannot store the excess amino acids formed from the digestion of proteins. The liver breaks down the excess amino acids and produces urea. Blood in the hepatic vein has a high concentration of urea than the blood in other vessels namely, hepatic portal vein and hepatic artery.

Liver receives its blood supply from two sources: the hepatic artery brings oxygenated blood from the heart, while the hepatic portal vein brings blood from the intestine and other abdominal organs. The blood is returned from the liver to the heart by the hepatic veins.

7.1.2 Formed Elements

Red blood cells/corpuscles (Erythrocytes), white blood cells/corpuscles (Leucocytes) and platelets are collectively called formed elements (Thrombocytes).

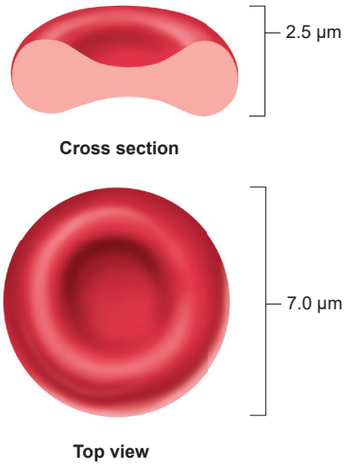


Figure 7.1 Structure of RBC

Red Blood Cells

Red blood cells are abundant than the other blood cells. There are about 5 million to 5.5 millions of RBC mm^{-3} of blood in a healthy man and 4.5-5.0 millions of RBC mm^{-3} in healthy women. The RBCs are very small with the diameter of about 7µm (micrometer). The structure of RBC is shown in Figure 7.1. The red colour of the RBC is due to the presence of a respiratory pigment, haemoglobin dissolved in the cytoplasm. Haemoglobin plays an important role in the transport of respiratory gases and facilitates the exchange of gases with the fluid outside the cell (tissue fluid).

The biconcave shaped RBCs increases the surface area to volume ratio, hence oxygen diffuses quickly in and out of the cell. The RBCs are devoid of nucleus, mitochondria, ribosomes and endoplasmic reticulum. The absence of these organelles accommodates more haemoglobin thereby maximising the oxygen carrying capacity of the cell.

The average life span of RBCs in a healthy individual is about 120 days after which they are destroyed in the spleen (graveyard / cemetery of RBCs) and the iron component returns to the bone marrow for reuse. **Erythropoietin** is a hormone secreted by the kidneys in response

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